50-trial search components (left two bars)
master LP solve stages (right bar)
Timeloop
Vizier + LP build
LP solve
Joint + hardware-screen solves
Fixed-chip controls
Branch and bound + certificate
EfficientNet (vision) — Vizier+LP speedup vs static
LLMs — Vizier+LP speedup vs static
151.0×
118.0×
140.7×
298.8×
197.1×
248.6×
312.0×
272.5×
204.3×
∞×
∞×
Hardware-search wall-clock (hours, log scale)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10.57h
4.2min
7.08h
3.6min
8.44h
3.6min
17.93h
3.6min
13.80h
4.2min
17.40h
4.2min
24.96h
4.8min
27.25h
0.10h
1.8min
47.00h
0.23h
1.6min
static infeasibl
1.17h
0.6min
static infeasibl
2.17h
0.53h
static
LLM
master
EffNet-b1
7.8M
EffNet-b2
9.2M
EffNet-b3
12M
EffNet-b4
19M
EffNet-b5
30M
EffNet-b6
43M
EffNet-b7
66M
Llama1-13B
13B
Llama1-65B
65B
Llama3-70B
70B
DeepSeek-V3
671B